

JANANI N N

Scientific Officer – Genomics & Molecular Biology

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PROFESSIONAL SUMMARY

Results-driven Biotechnology postgraduate with hands-on expertise in NGS library preparation, genomics assay development, and multi-platform molecular diagnostics. Experienced in DNA/RNA extraction from diverse sample types including FFPE, blood, bone marrow, tissue, and plasma; proficient in genotyping and gene expression profiling (GEP) workflows. Skilled in qPCR, ddPCR, RT-PCR, and Illumina NGS platforms with strong QC competence (Qubit, TapeStation, Nanodrop). Background in bioinformatics (RNA-Seq, WES, WGS, phylogenetics, molecular docking) and GCP-aligned documentation (ELN/LIS). Adept at cross-functional collaboration, SOP writing, and multi-project management – well-suited for translational medicine and clinical genomics programs.

CORE COMPETENCIES

NGS & Sequencing: Illumina NGS library prep (fragmentation, adapter ligation, PCR amplification, size selection), RNA-Seq, WES, WGS, genotyping, gene expression profiling (GEP)

PCR Platforms: RT-PCR, ddPCR, Overlap PCR, Nested PCR, Real-Time PCR, standard PCR, viral load testing – assay development & troubleshooting

Sample Handling: FFPE tissue, blood, bone marrow, plasma, saliva, liquid biopsy – DNA/RNA extraction (column-based & magnetic bead-based methods)

Nucleic Acid QC: Qubit fluorometry, Agilent TapeStation, Infinium assays, Nanodrop, NEB workflows

Molecular Biology: Gibson Assembly, molecular cloning, plasmid miniprep, cDNA synthesis, PBMCs isolation, ELISA, viral co-culture, transformation

Bioinformatics: Sequence analysis, phylogenetics, molecular docking, protein modelling, structural visualization, bioinformatics pipelines

Instruments: Qubit, Agilent TapeStation, CellSearch platform, Covaris, PCR systems, Nanodrop, HPLC, fluorescence & confocal microscopy

Data & Systems: LIS (sample registration, tracking, metadata), ELN, GCP-aligned documentation, MS Office, Google Workspace

PROFESSIONAL EXPERIENCE

Trainee Scientific Officer – Genomics | **Triesta Sciences, Bengaluru** | Apr 2025 – Present

- Perform DNA and RNA extraction from diverse sample types including FFPE tissue, fresh tissue, blood, bone marrow, and plasma using column-based and magnetic bead-based methods – directly applicable to translational medicine workflows.
- Execute nucleic acid QC using Qubit fluorometry, Agilent TapeStation, Infinium assays, and NEB workflows to ensure sample integrity for downstream genomic analysis.
- Conduct Illumina NGS library preparation covering fragmentation, adapter ligation, PCR amplification, and size selection; experienced in assay troubleshooting and run QC review.
- Manage sample registration, tracking, and metadata entry in the Laboratory Information System (LIS), ensuring data accuracy and traceability per GCP guidelines.
- Perform CTC (Circulating Tumor Cell) assays using the CellSearch platform, supporting liquid biopsy-based oncology diagnostics.
- Operate and maintain molecular biology instruments including Qubit, Agilent TapeStation, PCR systems, and Covaris; support AMC/CMC instrument maintenance activities.

Research Intern – Molecular Biology | **Jawaharlal Nehru Centre for Advanced Scientific Research (JNCASR), Bengaluru**
| Dec 2024 – Feb 2025

- Conducted HIV research using advanced PCR platforms including RT-PCR, ddPCR, Overlap PCR, Nested PCR, and Real-Time PCR; performed viral load testing on patient samples – strong foundation for clinical genotyping and qPCR-based assay development.
- Executed ELISA-based assays, DNA/RNA extraction, cDNA synthesis, and molecular typing for viral characterization in a BSL-2 research environment.

- Performed PBMCs isolation, viral co-culture, transformation, Gibson Assembly, gel elution, and plasmid miniprep for molecular cloning workflows.

EDUCATION

M.Sc. in Biotechnology | Kuvempu University, Shimoga, Karnataka | 2022 – 2024 | CGPA: 7.3 / 10

TECHNICAL SKILLS

Genomics: DNA/RNA extraction, Qubit & Nanodrop quantification, NGS library prep (Illumina), Agilent TapeStation, Infinium assays, gel electrophoresis, cDNA synthesis

PCR Techniques: RT-PCR, ddPCR, Overlap PCR, Nested PCR, Real-Time PCR, standard PCR, viral load testing

Molecular Biology: Gibson Assembly, molecular cloning, plasmid miniprep, transformation, PBMCs isolation, viral co-culture, molecular typing, ELISA

Proteomics: Protein purification (HPLC, Column Chromatography, TLC, SDS-PAGE), Western blotting, ELISA

Bioinformatics: Sequence analysis, phylogenetics, molecular docking, protein modelling, structural visualization, bioinformatics pipelines

Instruments: Qubit fluorometer, Agilent TapeStation, CellSearch platform, Covaris, PCR systems, Nanodrop, HPLC, fluorescence & confocal microscopy

IT & Data: MS Office, Google Workspace, LIS (sample management), data analysis tools

KEY PROJECT

Antioxidant Role of Lycopene and Lutein in Oxidative Damage Models

- Isolated and purified lycopene (tomato) and lutein (spinach) using HPLC and column chromatography.
- Investigated antioxidant properties against AAPH-induced oxidative damage in liposome and erythrocyte models.
- Assessed impact on erythrocyte ghost membrane protein profiles using SDS-PAGE analysis.

ALIGNMENT TO SYNGENE BBRC – ASSOCIATE SCIENTIST ROLE

- **Genotyping & GEP:** Direct experience with qPCR-based genotyping (ddPCR, RT-PCR, Nested/Overlap PCR) and NGS-based expression profiling from FFPE, blood, and liquid biopsy samples.
- **Assay Development & Qualification:** Hands-on assay execution, run QC review, and troubleshooting across NGS and PCR platforms – ready to support qualification/validation workflows.
- **FFPE & Liquid Biopsy:** Routine handling of FFPE tissue and blood/plasma samples for nucleic acid extraction and downstream genomics assays – exactly the sample types specified in the JD.
- **Bioinformatics:** Familiar with standard genomics/transcriptomics tools for data QC and interpretation (RNA-Seq, WES, WGS, sequence analysis, molecular docking).
- **GCP Documentation:** Trained in LIS/ELN record-keeping aligned with GCP guidelines at Triesta Sciences.
- **Cross-functional Collaboration:** Proven communication and project management skills across research and clinical genomics settings.

TRAINING & ACADEMIC ACTIVITIES

- National seminar on "Green Solutions – Exploring Pollution Management Strategies", Dept. of Environmental Science, Kuvempu University.
- Hands-on Workshop on Plant Tissue Culture, Dept. of Biotechnology, Kuvempu University.

LANGUAGES

English | Kannada | Hindi